II.1975 F L E A N E W S No. 4

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Please find herewith the fourth list of 1973 literature and the third one for 1974. Notifications of omissions will be appreciated.

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FIRST EUROPEAN CONFERENCE ON FLEAS

About 24 participants, from 11 European countries, are expected for the conference at Lund, Sweden, 3-5.VI.1975. Moreover, some of our colleagues from Canada and the U.S.A. also hope to attend. More detailed information is being sent from Lund to the prospective participants; may they all be able to make the journey !

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NOTES

Several colleagues kindly responded to my request, in Flea News (3), for copies of various 'obscure' publications and the receipt is gratefully acknowledged of papers by: Alvares & da Silva 1911; Bazex & Dupré 1965; Ducroiset 1968; Haiba 1962; Pessoa 1935.

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In my paper "On some adaptive structures in Siphonaptera" (1972, Folia Parasit. 19: 5-17) I described the anomalous condition, in a male of Rhadinopsylla integella, of "floating genitalia". Although there are several specimens of different species of fleas in the BMNH collection which are similarly abnormal, I would much like to see other additional material. If you have such male fleas with the 'phallosome-parameres package' as a separate unit inside the abdomen, i.e. not joined to tergum IX, could I borrow the specimen(s), please ?

F. G. A. M. Smit

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"In all ages wherein learning hath flourished, complaint hath been made of the itch of writing and the multitude of worthless books, wherewith importunate scribblers hath pestered the world."

J. Ray 1690

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ON USING A MICROSCOPE

Recently I heard an eminent acarologist remark on the excellence of numerous published and unpublished drawings of mites by the late Dr A. C. Oudemans (1858-1943) [the 'Ou' is pronounced as the 'ou' in e.g. louse or mouse, not as the 'oo' in e.g. loose or moose]. He added that among acarologists it has been said that Oudemans would never have required the use of a phase-contrast system as he must have had phase-contrast eyes.

In fleas, too, Oudemans observed structures (in the first decade of this century) which had presumably not been seen before and which are still extremely difficult to see with the ordinary light microscope (e.g., the transparent minute discs of the holding organs on the inner side of male antennae which have now been so well revealed by the scanning electron microscope).

Already in 1916 Dr H. Graf Vitzthum wrote that Oudemans was using superior optical aids, while in 1928 Dr A. Nalepa suggested that the Dutchman was able to examine objects at 2550x enlargement.

In an explanation of "Why I see so much more than anyone else" (1929, Ent. Ber., Amst. 7(167): 454-455) Oudemans stated that "... I have no superior optical aids, nor a microscope with 2550x magnification. I have an ordinary Leitz A 1 with ordinary objectives, all dating from 1895; the magnification of my oil-immersion system 1/16 with ocular 0 is only 530x linear. I need not say that with certain oculars one can get larger images but no better details... [In other words: the resolving power is in the objectives and e.g. for a 200x magnification 5x oc. + 40x obj. is better than 10x oc. + 20x obj. and is superior to 20x oc. + 10x obj.] The reason why I can see so much more than anyone else is simply that my microscope is placed in a so-called "Engelmann cabinet". This is 70 cm wide, 65 cm high and 35 cm deep and with the inside painted matt black. Therefore, when using the microscope, I am sitting in darkness; I can even draw two black curtains, which are fastened to the front (top) of the cabinet, over my back which then completely excludes day or artificial light. [Light, from behind the cabinet, was admitted to the mirror of the microscope through a matt black tube in the wall of the cabinet.] I advise those who want to have such a cabinet made to give it a greater height: too often I am bumping my skull against the upper side."

Although most of the microscopes in use today will have artificial illumination, they are still commonly occupying the place of the old-fashioned daylight/mirror type, namely in front of a window where it is difficult to cut down stray light when viewing. If a dark corner cannot be found for the instrument, it is advisable to slip e.g. a matt dark piece of cardboard over the oculars (± 20 x 30 cm, in which two circular holes the size and distance of the oculars). Or . . . make an Engelmann cabinet. One can make a large version of the cabinet by shielding off a corner of a room in which a black-topped table against a wall which has been painted matt black till at least one metre above the surface of the table.

When one looks around one may find that the microscope is still being misplaced and misused. Not everyone seems to realize the importance of:

(a) a good binocular microscope with a first-class optical system (because of the relative thickness of the specimens, phase-contrast is of no use in the case of mounted whole fleas)
(b) clean and dustfree equipment
(c) a well set up illumination (centring !)
(d) keeping the intensity of the light passing through the microscope as low as possible (one tends to use too strong a light)
(e) knowing how to use the substage condenser to its fullest advantage (especially a correct aperture, usually quite a small one)
(f) a blue (or green) filter
(g) a comfortable (adjustable) seat at a reasonably large table - both of the correct height for one's personal dimensions in order to avoid pain in the neck, back or elbow.
(h) the ability to observe and interpret (preferably correctly) -not merely to see- the observable. That seems to have been Oudemans' forte.

Has anyone other ideas, suggestions or comments on this subject ?

F. S.

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